Visualising GAP objects: current status and the future

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TODO: motivation

Currently there are multiple packages, which provide functionality to visualise GAP objects. First we will give an (possibly incomplete) overview of them with some examples.

XGAP (By Frank Celler and Max Neunhöffer)

Provides a GUI in which e.g. subgroup lattices can be investigated. Needs to be started separately.

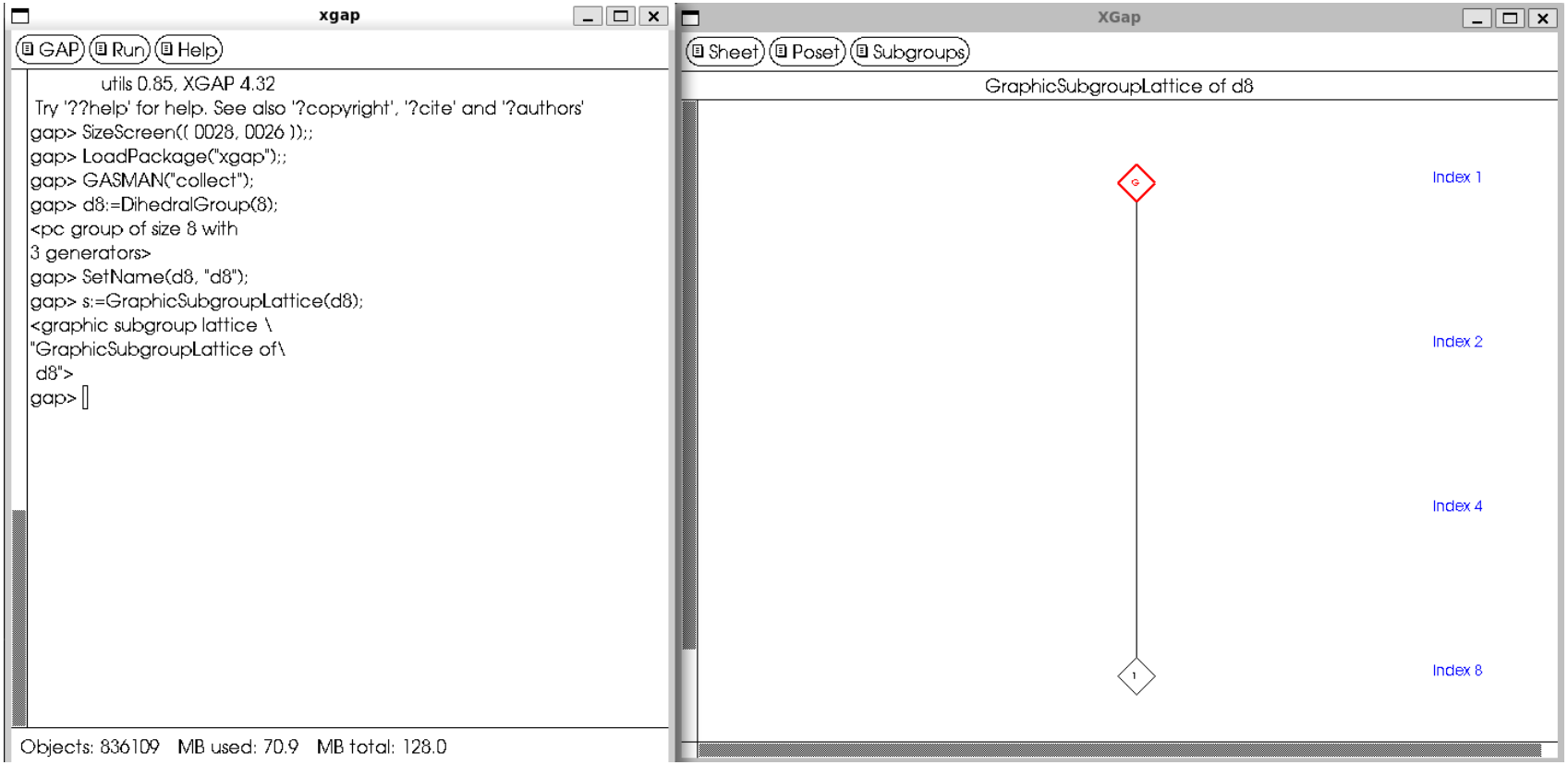
```
schnellecom@CURIE:/opt/gap/pkg/xgap$ ./xgap.sh
```

xgap

GAP Run Help

```
  GAP 4.14.0 of 2024-12-05
 GAP     https://www.gap-system.org
  Architecture: x86_64-pc-linux-gnu-default64-kv9
Configuration:  gmp 6.2.1, GASMAN
Loading the library and packages ...
Packages:   AClib 1.3.2, Alnuth 3.2.1, AtlasRep 2.1.9, AutPGrp 1.11,
            Browse 1.8.21, CaratInterface 2.3.7, CRISP 1.4.6, Cryst 4.1.29,
            CrystCat 1.1.10, CTblLib 1.3.9, curlInterface 2.4.0,
            FactInt 1.6.3, FGA 1.5.0, Forms 1.2.12, GAPDoc 1.6.7,
            genss 1.6.9, IO 4.9.1, IRREDSOL 1.4.4, LAGUNA 3.9.7, orb 4.9.1,
            Polenta 1.3.10, Polycyclic 2.16, PrimGrp 3.4.4, RadiRoot 2.9,
            recog 1.4.3, ResClasses 4.7.3, SmallGrp 1.5.4, Sophus 1.27,
            SpinSym 1.5.2, StandardFF 1.0, TomLib 1.2.11, TransGrp 3.6.5,
            utils 0.85, XGAP 4.32
Try '??help' for help. See also '?copyright', '?cite' and '?authors'
gap> SizeScreen(( 0028, 0026 ));;
gap> LoadPackage("xgap");;
gap> GASMAN("collect");
gap> 
```

Objects: 836109 MB used: 70.9 MB total: 128.0


xgap
GAP
Run
Help
utils 0.85, XGAP 4.32
Try '??help' for help. See also '?copyright', '?cite' and '?authors'
gap> SizeScreen((0028, 0026));;
gap> LoadPackage("xgap");;
gap> GASMAN("collect");
gap> d8:=DihedralGroup(8);
<pc group of size 8 with
3 generators>
gap> SetName(d8, "d8");
gap> s:=GraphicSubgroupLattice(d8);
<graphic subgroup lattice \
"GraphicSubgroupLattice of\
d8">
gap>
Objects: 836109 MB used: 70.9 MB total: 128.0
XGap
Sheet
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Subgroups
GraphicSubgroupLattice of d8
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Index 1
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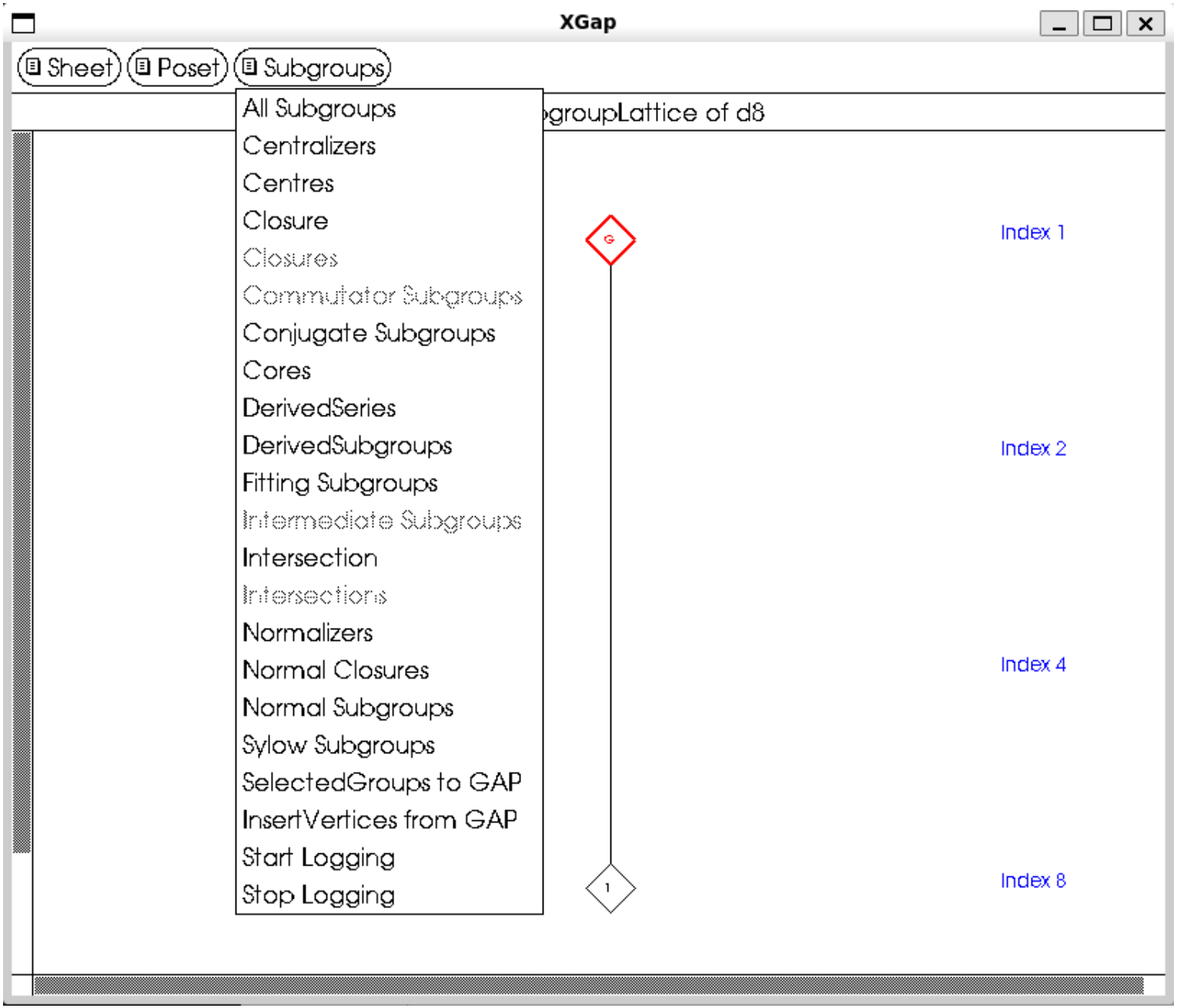
Index 8


XGap
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Subgroups
groupLattice of d8
All Subgroups
Centralizers
Centres
Closure
Closures
Commutator Subgroups
Conjugate Subgroups
Cores
DerivedSeries
DerivedSubgroups
Fitting Subgroups
Intermediate Subgroups
Intersection
Intersections
Normalizers
Normal Closures
Normal Subgroups
Sylow Subgroups
SelectedGroups to GAP
InsertVertices from GAP
Start Logging
Stop Logging
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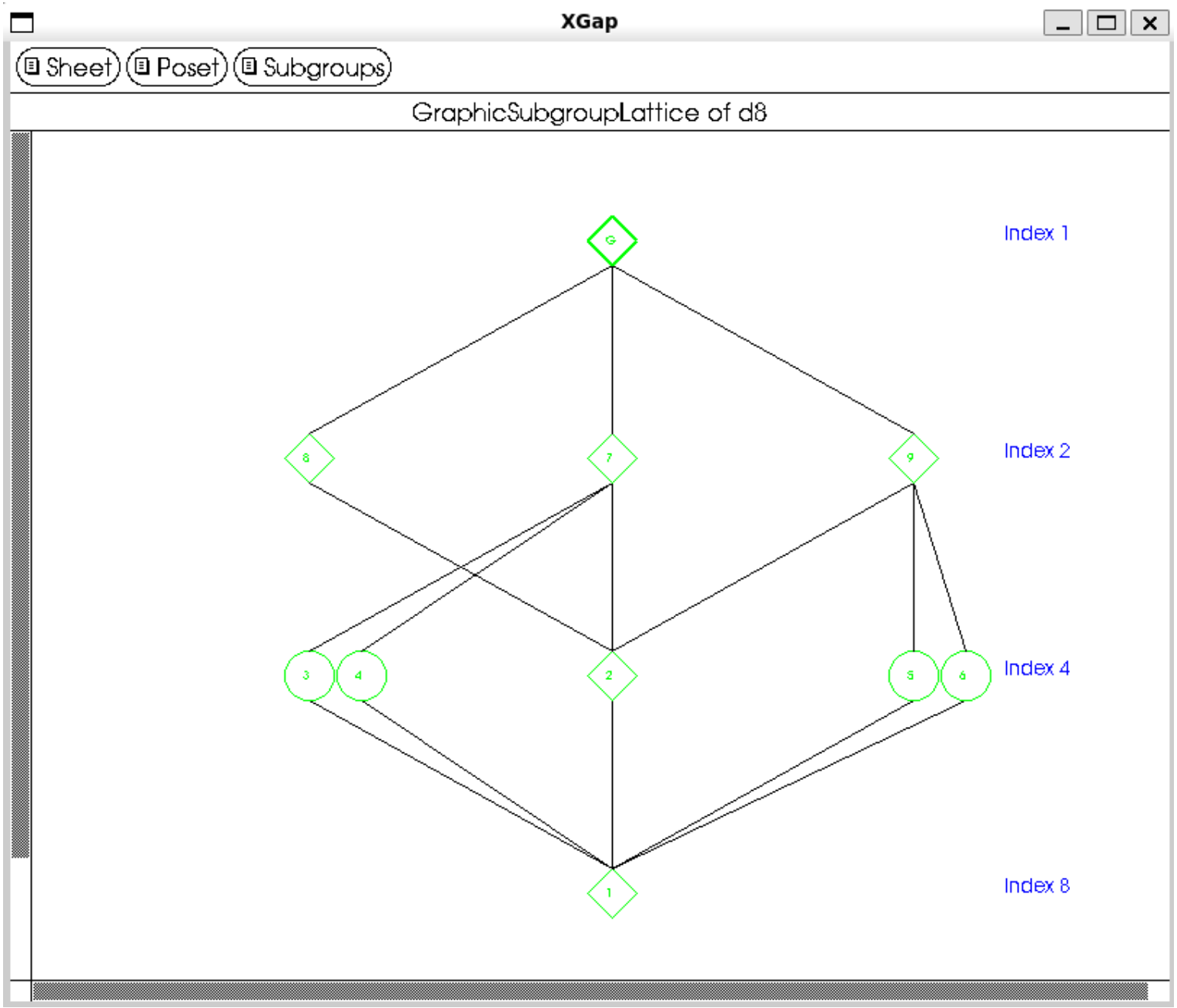
Index 8


XGap
Sheet
Poset
Subgroups
GraphicSubgroupLattice of d8
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1

Advantages

- Direct GUI to use
- Freely movable visualisation

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- Direct GUI to use
- Freely movable visualisation

Challenges

- Needs to be started standalone
- Needs to be adjusted to every OS

francy (by Manuel Martins)

Idea

Allow the XGAP system to run in a Jupyter Notebook.

Status

I was not able to get this running.

```
LoadPackage("digraphs"); LoadPackage("francy");

G := SymmetricGroup(3); as := AllSubgroups(G); nodes := [];
d := Digraph(as, {H, K} -> IsSubgroup(H, K));

vertices := DigraphVertices(d); edges := DigraphEdges(d);

canvas := Canvas(Concatenation("Subgroups Digraph of ",
  String(G)));
graph := Graph(GraphType.DIRECTED);
Add(canvas, graph);

customMessage := FrancyMessage(FrancyMessageType.INFO,
  "Simple Groups", "A group is simple if it is nontrivial
  and has no nontrivial normal subgroups.");

IsGroupSimple := function(i)
  Add(canvas, simpleGroupMessage);
  if IsSimpleGroup(as[i]) then
    Add(canvas, FrancyMessage("Simple",
      Concatenation("The vertex ", String(i),
        ", representing the subgroup ", String(as[i]),
        ", is simple.")));
  else
    Add(canvas, FrancyMessage("Not Simple",
      Concatenation("The vertex ", String(i),
        ", representing the subgroup ", String(as[i]),
        ", is not simple.")));
  fi;
  return Draw(canvas);
end;
```

```
for i in vertices do
  nodes[i] := Shape(ShapeType.CIRCLE, String(i));
  Add(nodes[i], Menu("Is this subgroup simple?",
      Callback(IsGroupSimple, [i])));
  Add(graph, nodes[i]);
od;

for i in edges do
  Add(graph, Link(nodes[i[1]], nodes[i[2]]));
od;

Draw(canvas);
```

(from [1])

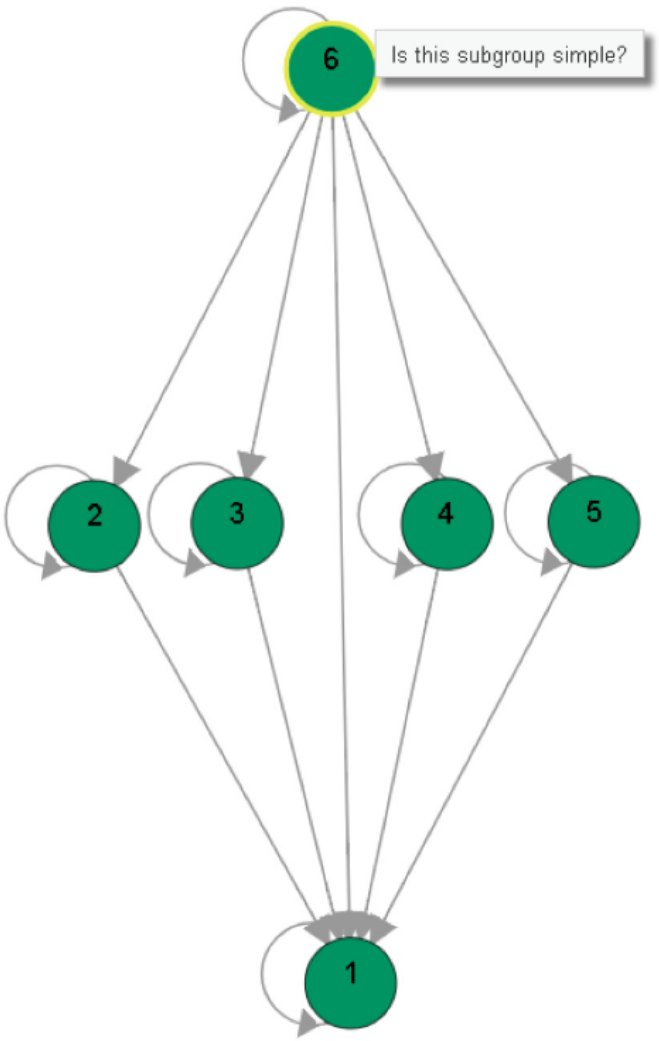
Francy

Subgroups Digraph of SymmetricGroup([1 .. 3])

Simple Groups A group is simple if it is nontrivial and has no nontrivial normal subgroups.

Simple The vertex 2, representing the subgroup Group([(2,3)]), is simple.

Not Simple The vertex 6, representing the subgroup SymmetricGroup([1 .. 3]), is not simple.



(from [1])

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- Runs in browser
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Other packages

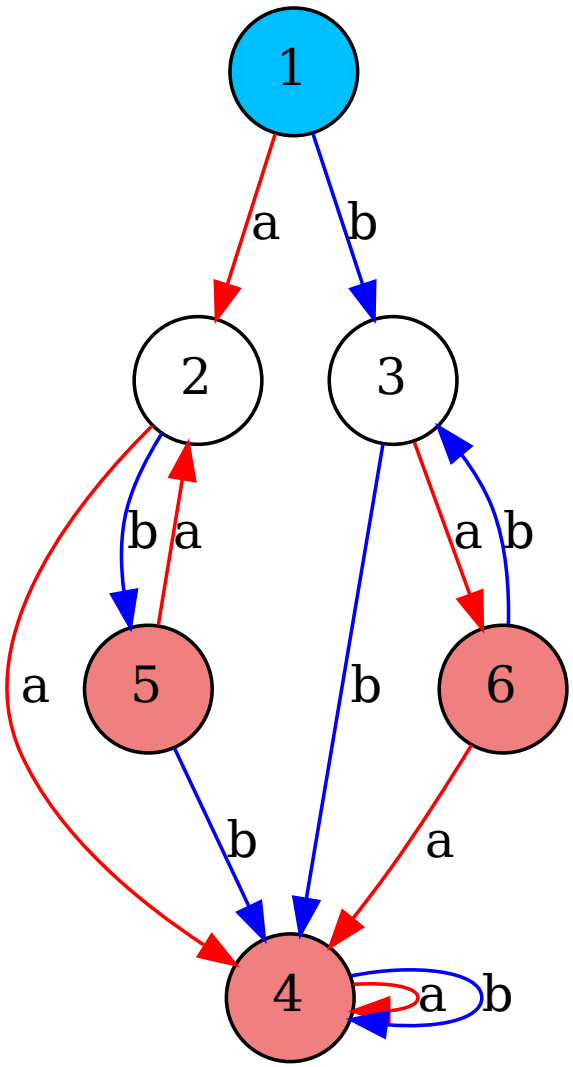
- jupyterviz
- semigroupviz

SgpViz (by Manuel Delgado and José João Morais)

Semigroup visualisation

- Can visualise e.g. Cayley graphs
- Using graphviz

```
gap> f := FreeMonoid("a","b");;
gap> a := GeneratorsOfMonoid( f )[ 1 ];;
gap> b := GeneratorsOfMonoid( f )[ 2 ];;
gap> r:=[[a^3,a^2], [a^2*b,a^2], [b*a^2,a^2], [b^2,a^2], [a*b*a,a], [b*a*b,b] ];;
gap> b21:= f/r;
<fp monoid on the generators [ a, b ]>
gap> DrawCayleyGraph(b21);
```


1
a
b
2
3
b
a
a
b
a
5
b
6
b
a
4
a
b

Advantages

- Uses graphviz standard
- Outputs PDFs

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Challenges

- Needs additional installation of graphviz and latex on system